

Employee Management System

1. Algorithm

Algorithm to Display Employee Details Using Constructor Overloading

Step 1: Start the program.

Step 2: Create an Employee class with the following data members:

- name
- id
- designation
- salary
- promotionStatus

Step 3: Define four constructors in the Employee class:

1. Default constructor.
2. Constructor with name and id.
3. Constructor with name, id, designation, and salary.
4. Constructor with all five employee details.

Step 4: Initialize the employee details according to the constructor being called.

Step 5: Define a display() method to display all employee details.

Step 6: In the main() method, create four employee objects:

- employee1 using the default constructor.
- employee2 using the two-parameter constructor.
- employee3 using the four-parameter constructor.
- employee4 using the five-parameter constructor.

Step 7: Display the heading **Employee Details**.

Step 8: Call the display() method for each employee object.

Step 9: Display the details of all four employees.

Step 10: Stop the program.

2. Execution Steps

Step 1: Create the Java Class

Create the class:

EmployeeManagement

inside the observation package.

Step 2: Create the Employee Class

The Employee class contains:

name

id

designation

salary

promotionStatus

Step 3: Execute the Default Constructor

```
Employee employee1 = new Employee();
```

The default values are:

Name : Not Assigned

Employee ID : 0

Designation : Not Assigned

Salary : 0.0

Promotion Status : Not Available

Step 4: Execute the Two-Parameter Constructor

```
Employee employee2 = new Employee("Rahul", 101);
```

The remaining values are assigned automatically:

Name : Rahul

Employee ID : 101

Designation : Employee

Salary : 0.0

Promotion Status : Not Available

Step 5: Execute the Four-Parameter Constructor

Employee employee3 =

```
new Employee("Priya", 102, "Manager", 65000);
```

Output:

Name : Priya

Employee ID : 102

Designation : Manager

Salary : 65000.0

Promotion Status : Not Promoted

Step 6: Execute the Five-Parameter Constructor

Employee employee4 =

```
new Employee("Arun", 103, "Senior Manager", 85000, "Promoted");
```

Output:

Name : Arun

Employee ID : 103

Designation : Senior Manager

Salary : 85000.0

Promotion Status : Promoted

Step 7: Call the display() Method

The display() method is called for all four objects:

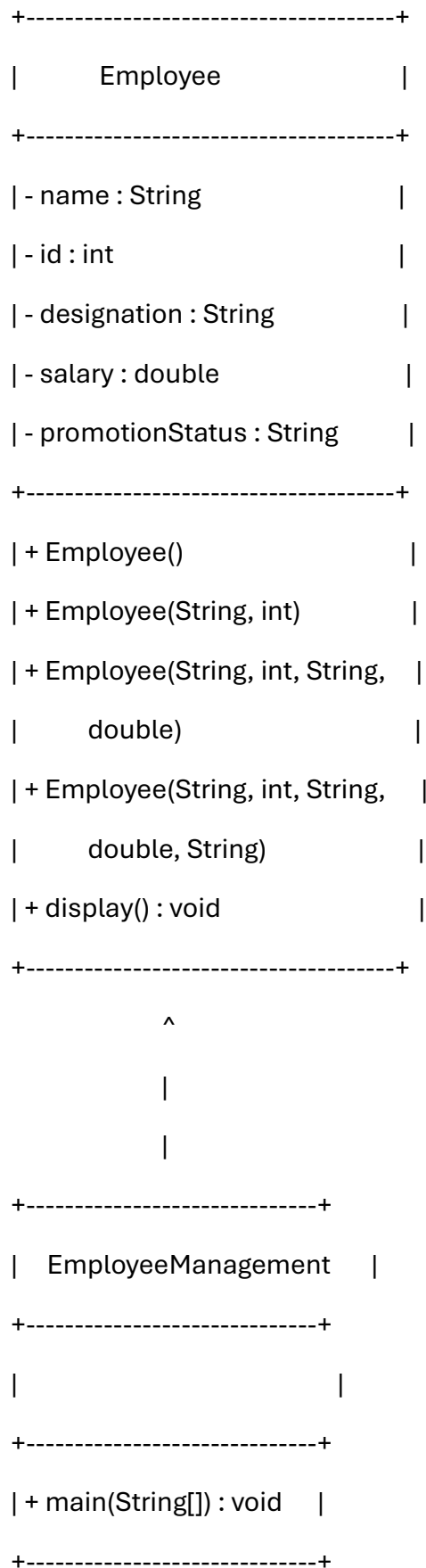
```
employee1.display();
```

```
employee2.display();
```

```
employee3.display();
```

```
employee4.display();
```

3. UML Class Diagram



Source Code:

```
package observation;

import java.util.Scanner;

class Employee {

    String name;

    int id;

    String designation;

    double salary;

    String promotionStatus;

    Employee() {

        name = "Not Assigned";

        id = 0;

        designation = "Not Assigned";

        salary = 0.0;

        promotionStatus = "Not Available";

    }

    Employee(String name, int id) {

        this.name = name;

        this.id = id;

        designation = "Employee";

        salary = 0.0;

        promotionStatus = "Not Available";

    }

    Employee(String name, int id, String designation, double salary) {

        this.name = name;
```

```
    this.id = id;

    this.designation = designation;

    this.salary = salary;

    promotionStatus = "Not Promoted";
}
```

```
Employee(String name, int id, String designation, double salary, String
promotionStatus) {

    this.name = name;

    this.id = id;

    this.designation = designation;

    this.salary = salary;

    this.promotionStatus = promotionStatus;
}
```

```
void display() {

    System.out.println("Name      : " + name);

    System.out.println("Employee ID   : " + id);

    System.out.println("Designation   : " + designation);

    System.out.println("Salary       : " + salary);

    System.out.println("Promotion Status : " + promotionStatus);

    System.out.println();

}

}
```

```
public class EmployeeManagement {

    public static void main(String[] args) {
```

```
Employee employee1 = new Employee();
```

```
Employee employee2 = new Employee(  
    "Rahul", 101  
);
```

```
Employee employee3 = new Employee(  
    "Priya", 102, "Manager", 65000  
);
```

```
Employee employee4 = new Employee(  
    "Arun", 103, "Senior Manager", 85000, "Promoted"  
);
```

```
System.out.println("Employee Details");  
System.out.println("-----");
```

```
employee1.display();  
employee2.display();  
employee3.display();  
employee4.display();
```

```
}
```

```
}
```

Output:

```
<terminated> EmployeeManagement (1) [Java Application] C:\Users\chinnu\OneDrive\Documents\eclipse-jee-2026-03-R-win32-x86_
Employee Details
-----
Name           : Not Assigned
Employee ID    : 0
Designation    : Not Assigned
Salary        : 0.0
Promotion Status : Not Available

Name           : Rahul
Employee ID    : 101
Designation    : Employee
Salary        : 0.0
Promotion Status : Not Available

Name           : Priya
Employee ID    : 102
Designation    : Manager
Salary        : 65000.0
Promotion Status : Not Promoted

Name           : Arun
Employee ID    : 103
Designation    : Senior Manager
Salary        : 85000.0
Promotion Status : Promoted
```